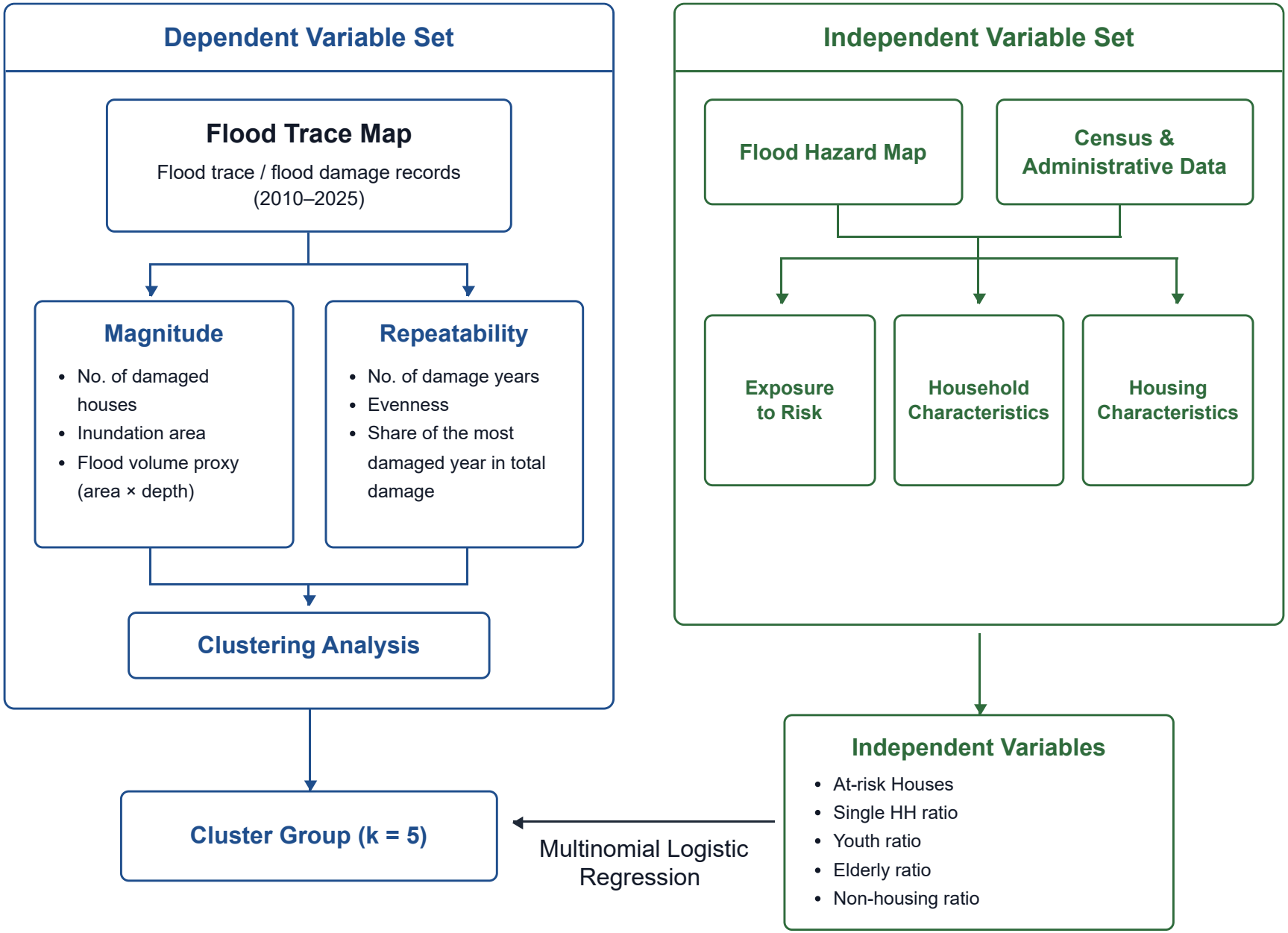

Dependent Variable Set
Flood Trace Map
Flood trace / flood damage records (2010–2025)
Magnitude
No. of damaged houses
Inundation area
Flood volume proxy (area × depth)
Repeatability
No. of damage years
Evenness
Share of the most damaged year in total damage
Clustering Analysis
Cluster Group (k = 5)
Independent Variable Set
Flood Hazard Map
Census & Administrative Data
Exposure to Risk
Household Characteristics
Housing Characteristics
Independent Variables
At-risk Houses
Single HH ratio
Youth ratio
Elderly ratio
Non-housing ratio
Multinomial Logistic Regression